# ZEN Univ.

# IUT-1 4th Ex.

**Def.**

$$\boxed{\begin{aligned} &\nu_p(n) \overset{\text{def}}{=} \text{exponent of prime } p \text{ of } n \\ &\text{e.g. } \nu_2(2^5 3^2) = 5. \\ &\qquad \nu_3(2^5 3^2) = 2 \end{aligned}}$$

1. (a) $\gcd(x-4,\ x+4)$

$$= \gcd\left(x-4,\ x+4-(x-4)\right)$$

$$= \gcd(x-4,\ 8) \mid 8$$

(b)

if $\nu_2(x-4) = 1$.

$x - 4 = 2^1 l \quad (\exists l \in \mathbb{N}^+, l \text{ is odd})$

$x + 4 = 2^1(l + 4)$

so $\nu_2(x+4) = 1$.

$(x-4)(x+4)$ can't be cubic. ※

if $\nu_2(X-4)=2$.

$\nu_2(X+4)=2$

So simillarly ※

Contradiction

if $\nu_2(X-4)=3$.

Only when $\nu_2(X+4)\equiv 0 \mod 3$,

$(X-4)(X+4)$ can be cubic.

if $v_2(X-4) \geq 4$,

$$v_2(X+4) = 3$$

So, only when $v_2(X-4) \equiv 0 \mod 3$,

$(X-4)(X+4)$ can be cubic.

if $v_2(x-4)=0$, $v_2(x+4)=0$.

Thus, $v_2(x-4) \equiv v_2(x+4) \equiv 0 \mod 3$

is necessity condition.

For any odd prime $p$, $p \mid (X-4) \Rightarrow p \nmid (X+4)$

by (a), and $p \mid (X+4) \Rightarrow p \nmid (X-4)$

as well.

So any odd prime $p \mid Y$, $v_p(Y^3) = v_p(X-4)$

or $v_p(Y^3) = v_p(X+4)$

finally, we got both 2 and odd prime

factors are cubical in $(X-4)$ and

$(X+4)$.

(C) From (b),

$$(X-4, X+4) = (-8, 0), (0, 8)$$

are only candidates. ⟨ Possibly non-trivial, by evaluating increasion of $x^3$

if $(-8, 0)$, $X = -4$, $Y = 0$

if $(0, 8)$, $X = 4$, $Y = 0$.

Finally, $(X, Y) = (\pm 4, 0)$ ∎

2. (a) $0 \in J \Rightarrow \underline{0+I \in \bar{J}}$

For any $a+I, b+I \in \bar{J}$,

$$(a+I)(b+I) = (ab+I)$$

$ab \in J$ so $\underline{ab+I \in \bar{J}}$

For any $x+I \in R/I$, $a+I \in \bar{J}$

$$(a+I)(x+I) = ax+I$$

$ax \in J$ so $\underline{ax+I \in \bar{J}}$

∎

(b) Consider $(a+I)(b+I) \in \overline{P}$

$$\begin{aligned} &\| \\ &ab+I \end{aligned}$$

$ab \in P$, and $P$ is prime then $\underline{a \in P \text{ or } b \in P}$

$\Downarrow$

$\underline{a+I \in \overline{P} \text{ or } b+I \in \overline{P}}$

$\blacksquare$

(c) $f: R \to R/I$.

$0+I \in Q$ then $\underline{0 \in f^{-1}(Q)}$.

$a, b \in f^{-1}(Q)$

$a+I = p+I$

$b+I = q+I$ for some $p+I, q+I \in Q$

$(p+I)(q+I) = (a+I)(b+I) = ab+I$

so $\underline{f^{-1}(Q) \ni ab}$

$a \in f^{-1}(Q)$, $(x+I)(a+I) = ax+I$, $\underline{f^{-1}(Q) \ni ax}$ so it's ideal.

Consider if $f^{-1}(Q) \ni ab$

$$f^{-1}(q+I) = ab$$

$$q + I = ab + I = (a+I)(b+I)$$

$Q$ is prime, so $a+I \in Q$ or $b+I \in Q$

$$\Downarrow \qquad\qquad \Downarrow$$

$f^{-1}(Q) \ni a$ or $f^{-1}(Q) \ni b$

So it's prime. ∎

Tool

Given $f, g$,

$\forall_x \; f \circ g(x) = x$

and

$\forall_y \; g \circ f(y) = y$

then $f$ and $g$ is bijection and

$\underline{f^{-1} = g}$

(d) $P \subseteq f^{-1}(\overline{P}))$

Given $x \in P$, $x+I \in \overline{P}$ then $x \in f^{-1}(\overline{P})$

$P \supseteq f^{-1}(\overline{P}))$

Given $x \in f^{-1}(\overline{P})$, $x+I \in \overline{P}$ then $\exists y \in P$, $x+I = y+I$.

here, $P \supseteq I$ so $x \in P$

$Q \subseteq \overline{f^{-1}(Q)}$ )

Given $x+I \in Q$, $x \in f^{-1}(Q)$ then $x+I \in \overline{f^{-1}(Q)}$

$Q \supseteq \overline{f^{-1}(Q)}$ )

Given $x+I \in \overline{f^{-1}(Q)}$, $\exists y \in f^{-1}(Q)$, $x+I = y+I \in Q$

$\blacksquare$

3. (a) $\overline{M}$ is ideal as well as 2-(a)

Assume $\overline{M}$ is not maximal. $\exists J$, $\overline{M} \subsetneq U \subsetneq R/I$

Take $a+I \in U \setminus \overline{M}$.

$a \notin M$ so $M + (a) = R$

This means any $x \in R$, $\exists m \in M, y \in (a)$. $x = m + y$.

$$x + I = (m + I) + (y + I) \in U$$

so $U = R$. $\times$ contradiction. $\blacksquare$

(b) Consider $f^{-1}(L)$ is not maximal.

$\exists\ J$ : ideal, $\quad f^{-1}(L) \subsetneq \underline{J \subsetneq R}$

Take any $\underline{a \in}\ \underline{J} \setminus \underline{f^{-1}(L)}$

$$f(J) \supseteq f\left(f^{-1}(L) \cup \{a\}\right) \supsetneq f\left(f^{-1}(L)\right) = L$$

non-trivial, but easy to check

not equal because

if $a + I \in f(f^{-1}(L))$

$\Rightarrow \exists\, b \in f^{-1}(L).\ a + I = b + I$

$\Rightarrow \underline{a \in f^{-1}(L)}$ ※ (by $f^{-1}(L) \supseteq I$)

$L$ is maximal, so $f(J) = R/I$ then $J = R$ ※ ■

(c) Same as 2-(a) and 3-(a),(b) ■

Tool

Consider under ZFC.

Zorn's lemma

non-empty subset totally ordered

Consider $P$ is non-empty poset, and any chain in $P$ has upper bound in $P$, then $P$ has at least one maximal element.

4. (a) Define $P$ as all proper ideals. $P \ni (0)$ so non-empty.

Take $\subseteq$ as partial order on it.

For any chain $C$ in $P$, $\cup C$ is upper bound.

$\because$ Any $I \in C$, $0 \in I$ so $0 \in \cup C$

For any $a, b \in \cup C$, $\exists I, J \in \cup C$, $a \in I$, $b \in J$.

$C$ is chain, so $\max(I, J) \ni a + b$.

For any $x \in R$, $a \in \cup C$, $\exists I \in C$, $a \in I$.

so $ax \in I \subseteq \cup C$.

Therefore $\cup C$ is ideal.

Any $I \in P$ doesn't have $1$, so $1 \notin \cup C$

then it's proper.

By applying Zorn's lemma,

There're maximal ideal in $P$.

(b) We can get this by replacing $P$ in proof of 4-(a)

as $P = \{ J : J \text{ is proper ideal and } J \supseteq I \}$ ∎

( and (a) is now special case as $I = (0)$ of (b) )

(c) $(a)$ is proper ideal by unity of $a$,

and by applying 3-(b), we can get

maximal ideal $M \supseteq (a) \ni a$ ∎